125101 - Analytical chemistry 1 for engineers

Tutorial 8 – Precipitation/Redox Titration

T8-1. Consider the titration of 50 mL KBr (0.1 M) with $AgNO_3$ (0.2 M):

$$Ag^+ + Br^- \longrightarrow AgBr_{(s)} \quad K_{sp}\,(AgBr) = 5.0\times10^{-13}$$

Find $p[Ag^+] = -log\,[Ag^+]$ for the following titrant volumes:

a. 15mL

b. 25mL

c. 30mL

Try to find the V_e of Ag^+ for this titration firstly,

as V_o of KBr = 50 mL, $[Br^-]_o$ = 0.1 M, $[Ag^+]$ = 0.2 M

50 mL × 0.1 M = V_e × 0.2 M ⟹ V_e = 25 mL

a. When V = 15 mL < V_e, which is before the equivalence point, there are excess of Br^- in solution.

$$[Br^-] = \frac{[Br^-]_o V_o - [Ag^+]V}{V_o + V} = \frac{0.1\times50-0.2\times15}{50+15} = 0.0308\ M$$

$$[Ag^+] = \frac{K_{sp}}{[Br^-]} = \frac{5.0\times10^{-13}}{0.0308} = 1.623\times10^{-11}\ M$$

$$p[Ag^+] = -\log(1.623\times10^{-11}) = 10.79$$

b. when V = 25 mL = V_e, it is at the equivalence point, the $p[Ag^+]$ can be obtained as,

$$p[Ag^+] = -\log\sqrt{K_{sp}} = -\log\sqrt{5.0\times10^{-13}} = 6.15$$

c. when V = 30 mL > V_e, it is the after the equivalence point, the $[Ag^+]$ is,

$$[Ag^+] = \frac{[Ag^+]V - [Br^-]_o V_o}{V_o + V} = \frac{0.2\times30-0.1\times50}{50+30} = 0.0125\ M$$

$$p[Ag^+] = -\log 0.0125 = 1.90$$

T8-2. What is the concentration of CrO_4^{2-} that should be added to solution in Mohr titration of the followings halides with Ag^+?

Ksp (AgCl) = 1.8×10^{-10}

Ksp (AgBr) = 5.0×10^{-13}

Ksp (AgI) = 8.3×10^{-17}

Ksp (Ag_2CrO_4) = 1.2×10^{-12}

At equivalence point, [Ag+] = $\sqrt{K_{sp}(AgI)}$ = 9.11×10^{-9}

Ksp (Ag_2CrO_4) = $[Ag^+]^2[CrO_4^{2-}]$

$\Longrightarrow [CrO_4^{2-}]$= Ksp ($Ag_2CrO_4$) / $[Ag^+]^2$=$1.2\times10^{-12}/(9.11\times10^{-9})^2$= 1.5×10^4 M

So, the indicator of $[CrO_4^{2-}]$ for I^- titrated with Ag^+ is 1.5×10^4 M;

The same idea for the calculation of another halides,

The indicator of $[CrO_4^{2-}]$ for Br^- titrated with Ag^+ is 2.4 M;

The indicator of $[CrO_4^{2-}]$ for Cl^- titrated with Ag^+ is 0.0071 M;

Therefore, it asks for higher the concentration of the indicator $[CrO_4^{2-}]$ when the K_{sp} is less in Mohr titration.

T8-3. 30 mL of a solution containing an unknown concentration of Cl^- was added to 50 mL of $AgNO_3$ with a concentration of 0.365 M. Later, the precipitate, $AgCl$, was filtered from the solution. The indicator, Fe^{3+}, was added to the filtered solution and the solution was titrated with KSCN with a concentration of 0.287 M. After the addition of 37.6 mL of the titrant, the solution turned to a pale rose color. What is the concentration of the Cl^- in the pristine solution?

The reactions:

$$Ag^+ + Cl^- \rightarrow AgCl(S) \quad K_{sp}(AgCl)= 1.8 \times 10^{-10}$$

The excess Ag^+ was titrated by KSCN:

$$Ag^+ + SCN^- \rightarrow AgSCN(S) \quad K_{sp}(AgSCN)= 1.2 \times 10^{-12}$$

As K_{sp}(AgCl) > K_{sp}(AgSCN), which means AgCl is more soluble, this is the reason why the precipitate AgCl was necessary to be filtered from the solution. If not, another reaction will occur:

$$AgCl + SCN^- \rightarrow AgSCN(S) + Cl^-$$

In this case, the quantity of SCN^- would become larger.

The quantity Ag^+ was added in total:

- Total (n_{Ag+}) = 0.365 mol/L × 0.05 L = 0.01825 mol

Now, let's calculate how many mols were titrated by SCN^-:

- Excess (n_{Ag+}) = n_{SCN-} = 0.287 mol/L × 0.0376 L = 0.01079 mol

So, the moles of Ag^+ reacts with Cl^- is,

- React with Cl^- (n_{Ag+}) = Total (n_{Ag+}) - Excess (n_{Ag+}) = 7.46×10^{-3} mol

And the $[Cl^-] = \frac{7.46\times10^{-3}\ \text{mol}}{0.03\ \text{L}}$ = 0.249 M

If the precipitant of AgCl was forgot to filter from the solution, the quantity of SCN^- would become larger. That's to say, the amount of Excess (n_{Ag+}) ↑. The final result for Cl^- will be less,

React with Cl^- (n_{Ag+}) ↓ = Total (n_{Ag+}) - Excess (n_{Ag+}) ↑